BINOMIAL PROCESSES

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Key concepts:

- Beta distribution
- Conjugate prior
- The Proportionality Trick

1. Introduction

We have seen that a Bayesian analysis depends on two critical ingredients:

- A *model* which describes the probability distribution of the data y as a function of an unknown parameter θ. This is the *likelihood function* and will be denoted by $p(y|\theta)$, where y denotes the data.
- A *prior distribution* for θ that describes our (subjective) beliefs about θ before we had observed the data y. We will denote the prior by $p(\theta)$.

Bayes' theorem tells us how to combine these two components to obtain the *posterior distribution* of the parameter θ, denoted $p(\theta|y)$, which summarizes what our beliefs about θ should be after observing the data y.

Today, we consider a different model—the binomial—and various priors for the parameter of this model. We discuss various ways to compute the posterior distribution: analytically, numerical quadrature, and independent and Markov Chain Monte Carlo (MCMC). This model is extremely simple (a single data value and a single parameter), but, unlike the examples from the first lecture, illustrates the standard way Bayesian analysis is actually done.

The binomial process is a model for *Simple Random Sampling with Replacement* (SRSWR) for a dichotomous attribute. At each draw, we select a unit from the population,[1] record whether the selected unit has the attribute or not, and return it to the population ("with replacement"). We repeat the process n times (independently), with y denoting the total number of successes in n draws.

To summarize the main results today:

[1] Selection may be made with either equal or unequal probability. The process will still be Binomial so long as the draws are independent and the probabilities don't change between the draws. If the selection probabilities are the same for all population units, then the parameter θ of the Binomial distribution will be the proportion of persons in the population having the attribute. If the selection probabilities are unequal, then θ is the ratio of the sum of the selection probabilities of those with the attribute to the sum of all selection probabilities. This has relevance for so-called "non-probability" sampling.

- If the prior distribution belongs to the Beta family (to be explained below), then the posterior distribution is also Beta. That is, the Beta distribution is *conjugate* to the Binomial likelihood function.
- The Uniform distribution is a special case of the Beta distribution.
- The Beta prior is interpretable as having observed a prior sample of data with a given number of successes and failures. The sum of the prior successes and failures is referred to as "the number of prior-equivalent observations." The posterior parameters are obtained by adding the number of successes and failures in the data to the prior number of successes and failures.
- The posterior mean of the success probability is a weighted average of the prior mean and the posterior mean. The posterior variance is always smaller than the prior variance.

Some of these results hold for other models, but others are specific to this particular model.

2. Bayesian Analysis of a Binomial Observation

2.1. **Binomial Model and Likelihood Function.** The Binomial model for y, the number of successes in n independent Bernoulli trials with success probability θ, is

$$y \sim \text{Bernoulli}(n, \theta).$$

The associated likelihood function is

$$p(y|\theta) = \binom{n}{y} \theta^y (1-\theta)^{n-y}, \qquad (y = 0, \dots, n)$$

We take n (the "number of trials") as known, so there is a single unknown parameter θ satisfying $0 \leq \theta \leq 1$. (That is, the *parameter space* is the closed interval $[0, 1]$.)

2.2. **Uniform Prior.** We start with the simplest possible prior for the success probability: $\theta \sim \text{Uniform}(0, 1)$. This is, in fact, the prior Bayes' himself used in his 1763 paper, "An Essay towards solving a Problem in the Doctrine of Chances," *Philosophical Transactions of the Royal Society of London*, vol. 53 (1763), pp. 370–418. Bayes thought that this prior distribution corresponded to "nothing being known" about θ.

$$p(\theta) = \begin{cases} 1 & \text{for } 0 \leq \theta \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Since θ is a probability, $p(\theta)$ is a probability distribution over a probability, which makes no sense with objective probabilities. For now, let's skirt the philosophical issues and see what we can learn about the unknown probability θ.

2.3. **Posterior.** We shall use Bayes' Theorem in the following form:

$$p(\theta|y) = \frac{p(\theta)p(\theta|\theta)}{p(y)} = \frac{p(\theta)\, p(y|\theta)}{\displaystyle\int p(\theta)\, p(y|\theta)\, d\theta} = \frac{\binom{n}{y}\theta^y(1-\theta)^{n-y}}{\displaystyle\int_0^1 \binom{n}{y} t^y (1-t)^{n-y}\, dt} = \frac{\theta^y(1-\theta)^{n-y}}{\displaystyle\int_0^1 t^y(1-t)^{n-y}\, dt}.$$

Bayes' did not know how to compute the integral in the denominator of the last term, but it is easy to see that $p(\theta|y)$ is a probability density function:

- Plainly, both the numerator and denominator are non-negative for $0 \leq \theta \leq 1$.
- The area under $p(\theta|y)$ equals one since the denominator is just the integral of the expression in the numerator.

In fact, this distribution has a name: the *Beta distribution*. We take a slight detour to discuss the Beta distribution and some other interesting stuff in the next section for those who want more detail.

Example 1. *Suppose we draw a random sample of size* $n = 5$ *using SRSWR and* $y = 4$ *of the five selected are Democrats. Let* θ *denote the proportion of Democrats in the population and suppose our prior beliefs are* $\theta \sim$ *Uniform*$(0, 1)$. *The prior and posterior distributions are shown below.*

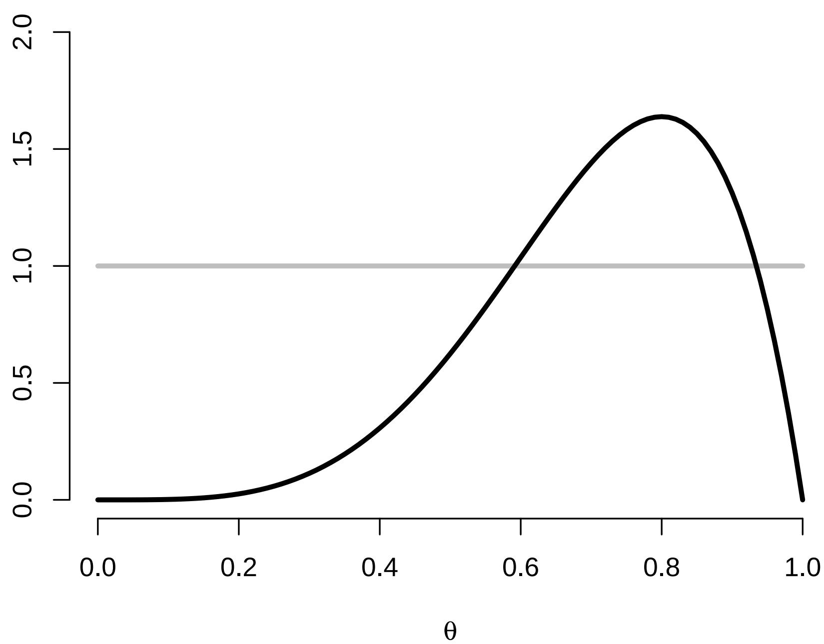

FIGURE 1. The Uniform$(0, 1)$ prior distribution for θ is shown in gray and the posterior Beta$(5, 2)$ distribution is shown in black. With a uniform prior, the posterior distribution is proportional to the likelihood function.

The R code to produce Figure 1 is shown below.

```
> curve(dunif(x, min=0, max=1), from=0, to=1, xlim=c(0,1), ylim=c(0,2),
+   bty="n", xlab=expression(theta), ylab="", main="", col="grey", lwd=3)
> curve(dbinom(y, size=n, prob=x)/beta(y,n-y), from=0, to=1, add=TRUE,
```

```
+   lwd=3, col="black")
```

3. Some Digressions

3.1. An Historical Note. Bayes did not know how to solve the integral in the denominator of $p(\theta|y)$, but found a beautiful argument to show that $p(y) = 1/(n+1)$ without solving the integral. He reasoned as follows. Consider a billiard table of unit length and roll $n+1$ billiard balls at random to points $x_1, \ldots, x_{n+1}$ in the unit interval $[0,1]$. Let $\theta = x_1$ (so $\theta \sim \text{Uniform}(0,1)$) and then let y denote how many of the remaining n balls are to the left of the first, *i.e.* the number of i for which $x_i \le x_1$ ($i \ge 2$). The locations of the balls are independent of one another and, given the position of the first ball, the probability that any other ball is to its left is

$$\mathbb{P}\{x_i \le x_1 | x_1\} = x_1 = \theta \qquad \text{for } i = 2, \ldots, n+1.$$

Thus, conditional upon θ, y is the number of successes in n Bernoulli trials, each with success probability θ, or $y \sim \text{Binomial}(n, \theta)$.

So far, so good: $\theta \sim \text{Uniform}(0,1)$ and, conditional on θ, $y \sim \text{Binomial}(n, \theta)$. But what is the marginal distribution of y? Using the physical analogy of the billiard balls, Bayes observed that an equivalent experiment would be to roll all $n+1$ balls and to pick one at random as θ. There are then $n+1$ possibilities (the selected ball may be to the left of all the balls, to the left of all but one, *etc.*) and each of these outcomes is equally likely. Therefore,

$$p(y) = \frac{1}{n+1} \qquad \text{for } y = 0, \ldots, n+1.$$

Pretty clever, no?

3.2. The Beta–Gamma Calculus. Today, we would use results on so-called *special functions* to do these calculations. The *Beta function* is defined to be

$$B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}\,dx \qquad (x, y > 0)$$

There is no closed form solution for $B(x,y)$, but you can use `beta(x,y)` to compute it in R.

The Beta function is closely related to another special function, the *Gamma function*, which we will also encounter repeatedly. For $x > 0$, define

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t}\,dt \qquad (x > 0)$$

In R, use `gamma(x)`.

Some useful properties of the Gamma function are:

- $\Gamma(1) = 1$ (obvious)
- $\Gamma(x+1) = x\Gamma(x)$ (integration by parts)
- For positive integers n, $\Gamma(n) = (n-1)!$ (by induction)

For more details, see Rudin, *Principles of Mathematical Analysis*, 3rd ed. (1976), pp. 192–195. The Gamma function extends the factorial function for non-integer arguments.

The relationship between the Beta and Gamma functions is given by the next result. (The proof is given for completeness and is not important.)

Proposition 1. *For $x, y > 0$,*

$$B(x,y) = \frac{\Gamma(x)\,\Gamma(y)}{\Gamma(x+y)}.$$

Proof.

$$\begin{aligned}
\Gamma(x)\,\Gamma(y) &= \int_0^\infty u^{x-1}e^{-u}\,du \int_0^\infty v^{y-1}e^{-v}\,dv = \int_0^\infty \int_0^\infty u^{x-1}v^{y-1}\,du\,dv \\
&= \int_0^\infty \int_0^{\pi/2} (r\cos^2\theta)^{x-1}(r\sin^2\theta)^{y-1}\,2r\sin\theta\cos\theta\,dr\,d\theta \\
&= 2\int_0^\infty r^{x+y-1}e^{-r}\,dr \int_0^{\pi/2} (\cos^2\theta)^{x-1}(\sin^2\theta)^{y-1}\,d\theta \\
&= \Gamma(x+y)\int_0^1 u^{x-1}(1-u)^{y-1}\,du = \Gamma(x+y)\,B(x,y),
\end{aligned}$$

and rearranging gives the desired result. In the second line, we made the substitutions $u = r\cos^2(\theta)$ and $v = r\sin^2(\theta)$ with Jacobian

$$\begin{vmatrix} \cos^2(\theta) & -2r\cos(\theta)\sin(\theta) \\ \sin^2(\theta) & 2r\sin(\theta)\cos(\theta) \end{vmatrix} = 2r\sin(\theta)\cos(\theta).$$

and in the last line $u = \cos^2(\theta)$ with $du = -2\cos(\theta)\sin\theta\,d\theta$. □

Using the preceding results, we see that for positive integers m and n,

$$B(m,n) = \frac{\Gamma(m)\,\Gamma(n)}{\Gamma(m+n)} = \frac{(m-1)!\,(n-1)!}{(m+n-1)!} = \frac{m+n}{mn}\frac{m!\,n!}{(m+n)!} = \frac{m+n}{mn}\binom{m+n}{m},$$

so the Beta function is closely related to the Binomial coefficients.

3.3. **The Beta Distribution.** A continuous random variable x is said to have the *Beta distribution* with parameters a and b when its density function is

$$p(x) = \frac{1}{B(a,b)}x^{a-1}(1-x)^{b-1} \qquad (0 \le x \le 1)$$

Clearly, $p(x) \ge 0$ for $0 \le x \le 1$ and

$$\int_0^1 p(x)\,dx = \frac{1}{B(a,b)}\int_0^1 x^{a-1}(1-x)^{b-1}\,dx = 1,$$

so $p(x)$ is a pdf with support on the interval $[0,1]$. We write $x \sim \text{Beta}(a,b)$ to indicate x has this distribution. The Beta pdf is programmed into R as `dbeta(x, shape1=a, shape2=b)`.

The Beta$(1,1)$ distribution is the same as the Uniform$(0,1)$ distribution. Further, it can be shown that if $X \sim$ Beta(a,b), then

$$\mathbb{E}(X) = \frac{a}{a+b} \qquad \mathbb{V}(X) = \frac{ab}{(a+b)^2(a+b+1)}.$$

The Beta is a family of continuous distributions on the unit interval that can be used as priors for probabilities like θ. Some examples of the Beta distribution are shown in Figure 1 below.

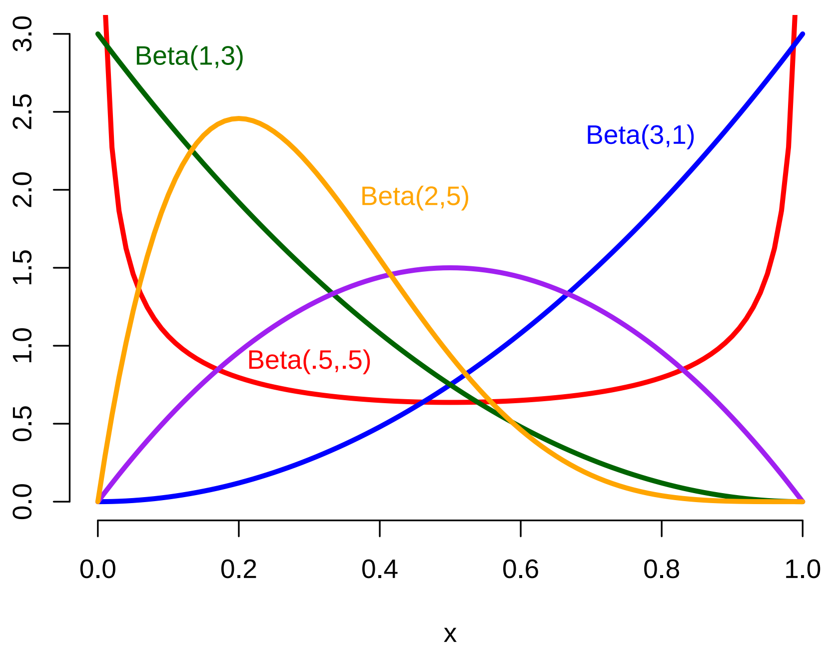

FIGURE 2. The Beta(a,b) distribution can be unimodal or bimodal (or uniform, as with the Beta$(1,1)$ distribution, not shown), depending upon the values of the shape parameters a and b.

4. CONJUGATE PRIORS

In the preceding example, the both the prior and posterior distributions belonged to the Beta family. (The prior was a Uniform$(0,1)$ distribution which is also a Beta$(1,1)$ distribution.) We will show that this holds when the prior is any Beta distribution (not just the Beta$(1,1)$ distribution). When the prior and posterior belong to the same family, then the prior distribution is said to be *conjugate* to the likelihood function.

Proposition 2. *If $\theta \sim$ Beta(a,b) and $y|\theta \sim$ Binomial(n,θ), then $\theta|y \sim$ Beta$(a+y, b+n-y)$.*

We discuss this result here, but defer the proof until Section 5.

First, the prior and posterior mean of θ are, using the result from Section 3.3, is

$$\mathbb{E}(\theta) = \frac{a}{a+b} \qquad \mathbb{E}(\theta|y) = \frac{a+y}{a+b+n}$$

so there is a simple formula for Bayesian updating.

Second, the posterior mean of θ is a weighted average of the prior mean $\mathbb{E}(\theta)$ and y/n ("the sample mean"):

$$\mathbb{E}(\theta|y) = \frac{a+b}{a+b+n}\frac{a}{a+b} + \frac{n}{a+b+n}\frac{y}{n} = w\mathbb{E}(\theta) + (1-w)\frac{y}{n},$$

where the weight on the prior mean is $w = (a+b)/(a+b+n) = n_0/(n_0+n)$ where $n_0 = a+b$ is the number of *prior equivalent observations*.

Third, the prior variance is (using the result from Section 3.3)

$$\mathbb{V}(\theta) = \frac{ab}{(a+b)(a+b+1)} = \frac{a}{a+b}\frac{b}{a+b}\frac{1}{n_0+1} = \frac{\mathbb{E}(\theta)[1-\mathbb{E}(\theta)]}{n_0+1},$$

which should remind you of the formula for the variance of a sample proportion. Similarly, the posterior variance is

$$\mathbb{V}(\theta|y) = \frac{\mathbb{E}(\theta|y)[1-\mathbb{E}(\theta|y)]}{n_0+n+1}.$$

Question: is the posterior variance always less than the prior variance?

Conjugate priors are a great convenience, especially for exponential families (like the binomial) which have a simple updating formula. Until the advent of Markov Chain Monte Carlo methods, it was difficult to do Bayesian analyses that did not involve conjugate priors. This was extremely limiting, but modern Bayesian computational methods have eliminated this constraint. Conjugate priors are still very useful as building blocks for more complex models.

5. The Proportionality Trick

It's easy to get lost in the calculations, but there is a trick that eliminates the need to perform the integration to calculate $p(y)$. Since $p(\theta|y)$ is a frequency function, it must sum or integrate to one. If we drop multiplicative terms that do not depend upon θ from the posterior distribution, we don't change the shape of the posterior. We can always recover the normalized posterior by summing or integrating and renormalizing. This device is frequently used in Bayesian statistics.

The *kernel* of a function of θ is any function obtained by dropping multiplicative terms that do not depend upon the argument θ. For example, we use the symbol "$\propto$" to mean "is proportional to" and write

$$p(y|\theta) \propto \theta^y(1-\theta)^{n-y}$$

to indicate that we have dropped the binomial coefficient $\binom{n}{y}$ from the kernel of the likelihood, since it does not depend upon θ.

Note that the prior predictive distribution $p(y)$ in the denominator of Bayes' Theorem *never* depends on θ, so we can write Bayes' Theorem as

$$p(\theta|y) \propto p(\theta)p(y|\theta)$$

by dropping the multiplicative term $1/p(y)$ from Bayes' Theorem. The product of the prior and likelihood is *not* a proper pdf, since it does not integrate to one. However, we can easily recover $p(\theta|y)$ by computing the integral and renormalizing,

$$p(\theta|y) = \frac{p(\theta)\, p(y|\theta)}{\displaystyle\int p(t)p(y|t)\, dt}$$

where the denominator is equal to the marginal distribution of y (by the Law of Total Probability). We try to avoid every having to do this integration. In most cases, it suffices to know that a normalizing constant exists and if we can recognize the form of the distribution, we won't have to actually calculate it.

In Bayesian analysis, we have three functions—the prior, likelihood, and posterior—which are functions of θ, each of which can often be simplified by dropping multiplicative terms not involving the parameter θ.

Name	**Function**	**Kernel**
Prior	$p(\theta) = \frac{1}{B(a,b)}\theta^{a-1}\theta^{b-1}$	$\theta^{a-1}(1-\theta)^{b-1}$
Likelihood	$p(y\|\theta) = \binom{n}{y}\theta^y(1-\theta)^{n-y}$	$\theta^y(1-\theta)^{n-y}$
Posterior	$p(\theta\|y) = \frac{1}{B(a+y,b+n-y)}\theta^y(1-\theta)^{n-y}$	$\theta^y(1-\theta)^{n-y}$

Table 1: Kernels of Functions in Binomial Bayesian Analysis

This gives a fairly simple recipe for Bayesian inference:

(1) Write the kernel of the prior $p(\theta)$, ignoring any multiplicative terms that do not depend upon θ. In this example, $p(\theta) \propto \theta^{a-1}(1-\theta)^{b-1}$.
(2) Write the kernel of the likelihood $p(y|\theta)$, again ignoring any multiplicative terms that do not depend upon θ. In this example, $p(y|\theta) \propto \theta^y(1-\theta)^{n-y}$.
(3) The posterior distribution of θ is proportional to the product of (the kernel of) the prior and the likelihood, $p(\theta|y) \propto p(\theta)p(y|\theta) \propto \theta^y(1-\theta)^{n-y}$.
(4) Either plot the kernel of the posterior to identify a distribution to which it belongs (since the kernel of a pdf uniquely identifies the pdf). In this example, $\theta^y(1-\theta)^{n-y}$ is the kernel of a Beta$(y+1, n-y+1)$ distribution.

In most cases, you needn't bother with finding the normalizing constant; knowing the shape of the posterior is good enough. We use this technique to prove Proposition 2 from Section 4 of these notes.

Proof of Proposition 2. Let us continue the example in the preceding section, except that we now allow the prior distribution for θ to be a Beta(a, b) distribution,

$$p(\theta) = \frac{1}{B(a,b)}\theta^{a-1}(1-\theta)^{b-1} \qquad (0 \le \theta \le 1)$$

This includes the Uniform$(0, 1)$ prior as a special case ($a = b = 1$). The kernels of these functions for the binomial model of the preceding section are shown in Table 1.

Bayes' Theorem states that

$$p(\theta|y) \propto p(\theta)p(y|\theta) \propto \theta^{a-1}(1-\theta)^{b-1}\theta^{y}(1-\theta)^{n-y} = \theta^{a+y-1}(1-\theta)^{b+n-y-1}$$

The integral of the unnormalized posterior is given by the Beta function

$$\int_0^1 \theta^{a+y-1}(1-\theta)^{b+n-y-1}\, d\theta = B(a+y, b+n-y)$$

so the posterior must be

$$p(y|\theta) = \frac{\theta^{a+y-1}(1-\theta)^{b+n-y-1}}{\int_0^1 \theta^{a+y-1}(1-\theta)^{b+n-y-1}\, d\theta} = \frac{1}{B(a+y, b+n-y)}\theta^{a+y-1}(1-\theta)^{b+n-y-1},$$

which is a Beta$(a + y, b + n - y)$ distribution. $\square$